AI 3000: Reinforcement Learning
Practice Problem Set 3

1 Stationary Distribution of a Markov Chain

Let P be the (one-step) transition probability matrix of a time-homogeneous Markov chain, i.e.

$$P(s' \mid s) = \mathbb{P}(S_{n+1} = s' \mid S_n = s)\,.$$

and P is row-stochastic.

(1.1) Show that the two step transition probability matrix is given by

$$\mathbb{P}(S_{n+2} = s' \mid S_n = s) = P^2(s' \mid s)$$

where $P^2 = P \cdot P$.

(1.2) Extend the above result by induction so that the k-step transition probability matrix is

$$\mathbb{P}(S_{n+k} = s' \mid S_n = s) = P^k(s' \mid s)$$

where $P^k = \underbrace{P \cdot \cdots \cdot P}_{k \text{ times}}$

(1.3) Let the initial distribution over the states be μ_0, i.e. $\mu_0 \colon \mathcal{S} \to [0,1]$, $\sum_{s \in \mathcal{S}} \mu_0(s) = 1$, and $S_0 \sim \mu_0$. We will identify μ_0 (and other pmfs) with a vector in $\mathbb{R}^{|\mathcal{S}|}$. Show that the marginal distribution μ_n of the Markov chain at time n (i.e. $S_n \sim \mu_n$) is given by

$$\mu_n^\top = \mu_0^\top P^n$$

(1.4) Show that the all ones vector $\mathbf{1}$ is an eigenvector of P with eigenvalue 1. Since P and $P^\top$ have the same eigenvalues, there exists some μ^* which is a left-eigenvector of P, i.e.

$$\mu^{*\top} = \mu^{*\top} P \quad \iff \quad \mu^* = P^\top \mu^*$$

(1.5) If $\mathcal{S}$ is finite, there exists such a μ^* with non-negative entries. Show that if you normalize such a μ^* so that its entries sum to 1 (and hence, can be interpreted as a pmf), then it is a stationary distribution of the Markov chain, i.e.

$$\mu_0 = \mu^* \implies \forall\, n \in \mathbb{N} : \mu_n = \mu^*$$

Remark We have not proved the existence of a stationary distribution here, nor its uniqueness, which only holds under further assumptions on P.

2 On span and sup-norm

Let $\mathrm{Sp}\colon \mathbb{R}^n \to [0,\infty)$ be the span seminorm on $\mathbb{R}^n$

$$\mathrm{Sp}(x) = \max x - \min x$$

where max and min are taken over the entries of $x \in \mathbb{R}^n$.

(2.1) Define the kernel of Sp as

$$\ker(\mathrm{Sp}) \coloneqq \{x \in \mathbb{R}^n \mid \mathrm{Sp}(x) = 0\}$$

Find $\ker(\mathrm{Sp})$ and show that it is a linear subspace of $\mathbb{R}^n$.

(2.2) Given a subspace X of $\mathbb{R}^n$, the projection onto X with respect to the sup-norm $\|\cdot\|_\infty$ is defined as

$$\Pi_X(y) \coloneqq \arg\min_{x\in X} \|y - x\|_\infty$$

and the distance to X is

$$\delta_X(y) \coloneqq \|y - \Pi_X(y)\| = \min_{x\in X} \|y - x\|_\infty$$

Show that

$$\delta_{\ker(\mathrm{Sp})}(y) = \frac{\max y - \min y}{2} = \frac{\mathrm{Sp}(y)}{2} \tag{1}$$

$$\Pi_{\ker(\mathrm{Sp})}(y) = \frac{\max y + \min y}{2} \tag{2}$$

Remark Equation (1) can be written as

$$\mathrm{Sp}(y) = \min_{x\in\ker(\mathrm{Sp})} 2\|y - x\|_\infty$$

This is a stronger version of the inequality we derived in the lectures

$$\mathrm{Sp}(y) \le 2\|y\|_\infty$$

3 Combinations of Policies

(Adapted from Puterman & Chan; Chapter 5, Exercise 13)
Consider an infinite-horizon MDP with discount factor $\lambda < 1$, finite state space $\mathcal{S}$ and finite action space $\mathcal{A}$. Let $\phi_1, \phi_2\colon \mathcal{S} \to \mathcal{A}$ be two deterministic decision rules, and $v^{\phi_1}_{\infty,\lambda}, v^{\phi_2}_{\infty,\lambda}$ be their respective value functions.

Define a new deterministic decision rule $\phi\colon \mathcal{S} \to \mathcal{A}$ as

$$\phi(s) = \begin{cases} \phi_1(s), & v^{\phi_1}_{\infty,\lambda}(s) \ge v^{\phi_2}_{\infty,\lambda}(s) \\ \phi_2(s), & v^{\phi_1}_{\infty,\lambda}(s) < v^{\phi_2}_{\infty,\lambda}(s). \end{cases}$$

and let $v^{\phi}_{\infty,\lambda}$ be its value function. Show that for every $s \in \mathcal{S}$,

$$v^{\phi}_{\infty,\lambda}(s) \ge \max\left\{v^{\phi_1}_{\infty,\lambda}(s), v^{\phi_2}_{\infty,\lambda}(s)\right\}.$$

4 Bellman Residual Minimization over an invariant family

Suppose we wish to approximate the optimal value function $v^*_{\infty,\lambda}$ of an infinite-horizon MDP with discount factor $\lambda < 1$, and maximum return operator $L\colon \mathbb{R}^{|\mathcal{S}|} \to \mathbb{R}^{|\mathcal{S}|}$ using a parametric family $\{v_\theta\}_{\theta\in\Theta}$. Assume that for any $\theta \in \Theta$, there is some $\phi \in \Theta$ such that

$$Lv_\theta = v_\phi\,.$$

Show that we can approximate $v^*_{\infty,\lambda}$ to any desired accuracy using $\{v_\theta\}_{\theta\in\Theta}$. Further, show that if $\{v_\theta\}_{\theta\in\Theta}$ is a closed set then $v^*_{\infty,\lambda}$ can be represented exactly.

Remark In particular, for a linear architecture, $v_\theta = \Phi\theta$ for some feature matrix Φ. The set $\{v_\theta\}_{\theta\in\Theta}$ is then the column space of Φ, which is closed.

5 Properties of linear regression

Consider an infinite-horizon MDP with discount factor $\lambda < 1$, maximum return operator L and optimal value function $v^*_{\infty,\lambda}$. Suppose we want to approximate $v^*_{\infty,\lambda}$ using a linear architecture with feature matrix B, and corresponding projection matrix $\Pi = B(B^\top B)^{-1}B^\top$. Show that

$$\arg\min_\beta \|B\beta - v^*_{\infty,\lambda}\|_2 = \arg\min_\beta \|B\beta - Lv^*_{\infty,\lambda}\|_2 = \arg\min_\beta \|B\beta - \Pi Lv^*_{\infty,\lambda}\|_2$$